

ROXAS, Philippines

WMO: 985380

Lat: 11.598N

Lon: 122.752E

Elev: 3

StdP: 101.29

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.8	24.2	20.8	15.5	25.8	21.5	16.2	25.9	10.1	25.1	9.1	25.9	3.7	40	0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	6.0	33.1	27.8	32.6	27.7	32.2	27.5	28.2	32.2	28.0	32.0	27.7	31.6	3.7	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	22.7	31.6	26.7	22.3	31.2	26.5	22.1	31.0	90.5	32.5	89.3	32.2	88.3	31.9	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	6.8	6.0	DB	22.2	35.3	1.8	1.9	20.9	36.6	19.8	37.7	18.8	38.8	17.5	40.2	(4)
(5)			WB	19.7	29.0	1.8	0.6	18.4	29.4	17.4	29.8	16.4	30.1	15.1	30.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.0	26.8	27.1	27.8	28.7	29.1	28.6	28.2	28.3	28.2	28.1	28.1	27.5	(6)	
	DBStd	1.12	0.92	0.83	0.94	0.91	0.98	0.99	0.94	0.85	0.92	0.90	0.92	0.99	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6587	521	478	552	562	593	557	563	568	544	562	543	543	(10)	
	CDD18.3	3545	263	244	293	312	334	307	305	309	294	304	293	285	(11)	
	CDH23.3	39893	2484	2427	3232	3850	4262	3637	3429	3474	3251	3458	3357	3031	(12)	
	CDH26.7	13384	443	527	977	1543	1874	1417	1196	1222	1107	1191	1102	786	(13)	
(14) Wind	WSAvg	2.9	4.1	3.6	3.4	3.0	2.4	2.2	2.1	2.2	2.1	2.6	3.5	4.2	(14)	
(15) Precipitation	PrecAvg	2315	120	89	90	67	157	266	281	184	217	308	280	255	(15)	
	PrecMax	4086	212	267	296	339	408	785	521	445	488	526	523	537	(16)	
	PrecMin	1001	52	27	19	7	22	68	104	53	57	93	105	95	(17)	
	PrecStd	613	45	67	76	86	107	142	113	97	119	112	116	122	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.7	31.0	32.0	32.9	34.0	33.5	33.1	33.1	33.1	32.6	32.2	31.7	(19)	
		MCWB	26.7	26.8	27.2	27.8	28.0	28.0	27.8	27.7	27.7	27.7	27.7	27.5	(20)	
	2%	DB	30.0	30.2	31.3	32.2	33.1	32.7	32.2	32.4	32.2	31.9	31.6	30.9	(21)	
		MCWB	26.5	26.4	26.9	27.5	27.9	27.8	27.4	27.4	27.4	27.4	27.4	27.1	(22)	
	5%	DB	29.4	29.7	30.7	31.8	32.6	32.2	31.7	31.8	31.6	31.4	31.1	30.3	(23)	
		MCWB	26.1	26.1	26.6	27.3	27.7	27.6	27.2	27.2	27.2	27.1	27.1	26.9	(24)	
	10%	DB	28.8	29.2	30.2	31.4	32.0	31.6	31.1	31.2	31.0	30.8	30.5	29.7	(25)	
		MCWB	25.8	25.8	26.3	27.0	27.5	27.3	27.0	27.0	27.0	26.9	26.9	26.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.2	27.3	27.7	28.2	28.6	28.5	28.2	28.2	28.1	28.1	28.1	27.7	(27)	
		MCDB	30.0	30.2	31.1	32.1	32.8	32.6	32.2	32.2	32.1	31.8	31.6	31.0	(28)	
	2%	WB	26.7	26.9	27.3	27.8	28.2	28.1	27.7	27.7	27.6	27.7	27.6	27.3	(29)	
		MCDB	29.5	29.7	30.7	31.7	32.4	32.2	31.6	31.7	31.6	31.3	31.1	30.5	(30)	
	5%	WB	26.3	26.5	27.0	27.5	27.9	27.7	27.4	27.4	27.3	27.4	27.3	27.0	(31)	
		MCDB	29.0	29.3	30.3	31.3	32.0	31.6	31.2	31.3	31.1	30.9	30.6	30.1	(32)	
	10%	WB	26.0	26.1	26.6	27.2	27.6	27.4	27.1	27.1	27.0	27.0	27.0	26.6	(33)	
		MCDB	28.5	28.8	29.8	30.9	31.6	31.2	30.7	30.8	30.7	30.5	30.2	29.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	3.7	4.2	4.8	5.4	6.0	6.1	5.8	6.0	5.9	5.5	4.8	4.1	(35)	
		MCDBR	4.6	4.9	5.4	5.8	6.6	6.8	6.6	6.7	6.7	6.2	5.6	5.0	(36)	
	5% WB	MCWBR	2.6	2.7	2.8	2.9	3.2	3.3	3.2	3.2	3.2	3.1	2.9	2.8	(37)	
		MCWBR	4.4	4.7	5.2	5.8	6.3	6.5	6.4	6.5	6.4	6.0	5.5	4.9	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.440	0.428	0.431	0.436	0.436	0.432	0.428	0.442	0.436	0.453	0.443	0.446	(40)	
		taud	2.391	2.417	2.416	2.402	2.425	2.447	2.459	2.408	2.421	2.365	2.395	2.373	(41)	
	Edn at Noon	Edn at Noon	865	890	890	876	861	854	860	859	872	854	855	849	(42)	
		Edh at Noon	119	120	122	123	117	113	113	120	119	124	118	119	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	4.06	4.96	5.65	6.19	5.65	5.12	4.72	4.82	4.82	4.66	4.51	3.94	(44)	
	RadStd	RadStd	0.45	0.51	0.57	0.34	0.51	0.36	0.40	0.32	0.46	0.31	0.36	0.30	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.37	+0.50	N/A	+0.46	+0.18	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		+0.16	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page